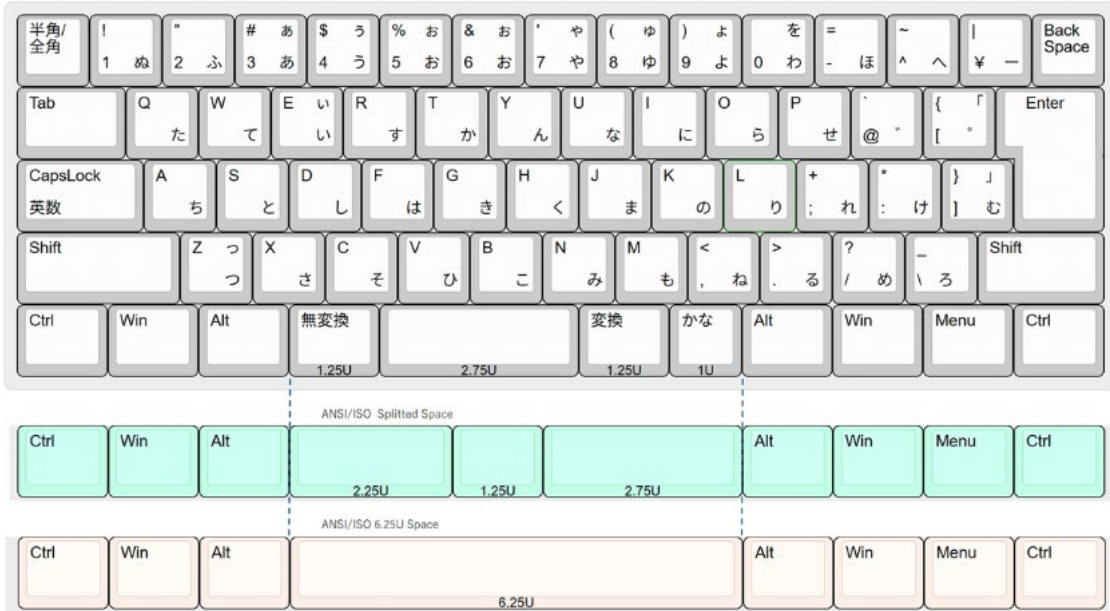


A Physical Design Proposal for Japanese JIS Keyboards

ISO-Connected JIS (IC-JIS)
Preserving Japanese input functions while reconnecting JIS design to reusable keyboard components



Executive Summary

Japanese JIS keyboards require locale-specific input keys, but this does not necessarily require product-specific physical dimensions throughout the keyboard. IC-JIS is a reference design that localizes JIS-specific differences while reusing established key widths, split-layout practices, and component ecosystems wherever technically feasible.

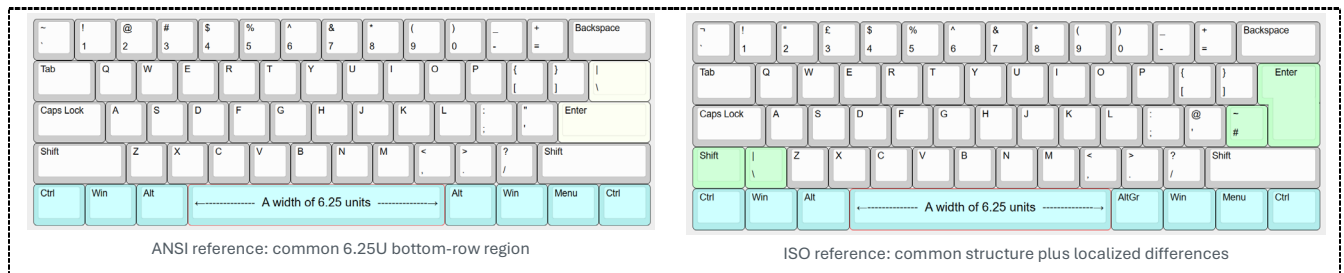
ISSUE	PROPOSAL	FIRST ASK
Bottom-row dimensions vary between products, affecting Japanese-input key area, replacement parts, and long-term serviceability.	Treat the common 6.25U space-bar region as a boundary and partition it using existing widths: 1.25U + 2.75U + 1.25U + 1U.	Do not decide on product adoption. Perform only a low-cost desktop layout feasibility check (Step 0).

Decision requested

The proposal does not claim that 2.75U is the only correct space-bar size. The goal is to make bottom-row trade-offs explicit, testable, and reusable across products.

1 Problem Statement and Design Opportunity

The central issue is not that a space bar is “too long.” The issue is whether the complete bottom row has an explainable balance among Japanese input usability, component availability, maintainability, and future reuse.



Design question

Recent JIS implementations include longer Space keys such as 3.75U, 4.25U, and 4.5U, sometimes alongside 1U Conversion / Non-Conversion keys. Enlarging one key necessarily changes the space available to the others.

Why this is worth reviewing

- Product-specific high-use dimensions may reduce replacement options after end of sale.
- A small JIS component market can be further fragmented by product-specific dimensions.
- A dimension adopted in one generation can propagate dedicated tooling, inventory, and design assets into later products.

2 Evidence from Historical JIS Implementations

Because public standards do not fully determine all physical key dimensions, historical products were compared using product photographs and ratios against standard 1U keys. Values not supported by official drawings are estimates rounded to 0.25U.

Examples around the OADG109A adoption period



Fujitsu FMV-KB323



Hitachi PC-KB5330



NEC PK-KB011



IBM Japan KB-8920

Current continuity example



Fujitsu FMV-KB338

3 Observed Design Pattern

Comparison of Bottom-Row Layouts on JIS Keyboards from Japanese Manufacturers

Manufacturer	Model number	Release year	Left Ctrl	Left Win	Left Alt	Non-Conversion 無変換 Muhenkan	Space	Conversion 変換 Henkan	KANA	Right Alt	Right Win	Menu /App	Right Ctrl
NEC	PK-KB011	Around 2000	1.5U	1.25U	1.25U	1.25U	2.75U	1.25U	1.25U	1.25U	1U	1U	1.25U
Fujitsu	FMV-KB323	Around 2000	1.5U	1.25U	1.25U	1.25U	2.75U	1.25U	1.25U	1.25U	1U	1U	1.25U
Hitachi	PC-KB5330	Around 2000	1.5U	1.25U	1.25U	1.25U	2.75U	1.25U	1.25U	1.25U	1U	1U	1.25U
IBM Japan	KB-8920	Around 2000	1.5U	1.25U	1.25U	1.5U	2.5U	1.5U	1.25U	1U	1U	1U	1.25U
Fujitsu	FMV-KB338	Current example	1.5U	1.25U	1.25U	1.5U	2.5U	1.5U	1.25U	1U	1U	1U	1.25U

Bottom-row comparison of the JIS keyboards reviewed in the source study



Observation

In the products reviewed, Space was approximately 2.5U to 2.75U, while Conversion and Non-Conversion were approximately 1.25U to 1.5U. The useful observation is not a “tradition of short space bars,” but a design tendency to reserve meaningful physical area for Japanese input keys.

This does not prove that 2.75U is inherently correct. A different design can be valid if its benefits, trade-offs, service strategy, and component implications can be explained and tested.

4 Design Objective and Evaluation Criteria

The objective is neither to standardize 2.75U as the only answer nor to replace JIS with ANSI. The objective is to preserve Japanese input functions while reducing unnecessary dependence on unique physical dimensions and reconnecting the design to reusable assets.



ANSI: a common component baseline



ISO: localized differences connected to a common baseline

Shift JIS implementation from “adding product-specific dimensions” toward “implementing limited locale-specific differences on a common design baseline.”

CRITERION	ENGINEERING QUESTION
Usability	Can Japanese input keys be identified and operated reliably by the thumbs?
Maintainability	Can worn or damaged parts be replaced after the original product lifecycle?
Ecosystem connectivity	Can existing widths, switches, stabilizers, and design practices be used?
Transferability	Can the rationale and design pattern be reused in another product generation?

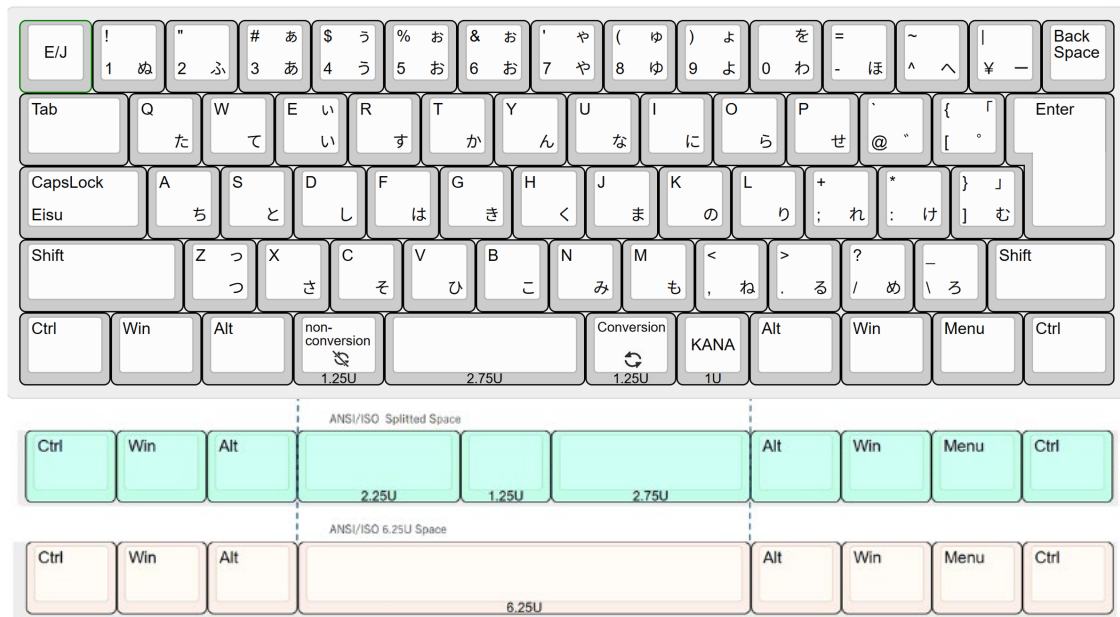
5 Reference Specification: ISO-Connected JIS (IC-JIS)

Use the region normally occupied by a 6.25U Space key as the boundary condition. Reserve widths first for the Japanese input keys, then allocate the remaining width to Space.



Dimension derivation

$6.25U - 1.25U \text{ Non-Conversion} - 1.25U \text{ Conversion} - 1.00U \text{ Katakana/Hiragana} = 2.75U \text{ Space}$



Relationship among the 6.25U region, split-space design, and the IC-JIS bottom row

Terminology

“ISO-Connected” is a design concept, not a standards classification. It describes reconnecting shared physical elements to the broader component ecosystem while retaining locale-specific JIS functions.

6 Implementation Architecture

JIS requires locale-specific switch positions, matrix mapping, keycodes, and legends. Those requirements do not automatically imply that every key width, stabilizer, or mechanical design pattern must form a separate ecosystem.



Reference implementation: split Right Shift in the PFU HHKB US layout



JIS application: the “Ro” key can be treated as the same split-right-Shift design problem

AREA	IC-JIS APPROACH	EXISTING ASSET TO VERIFY
Bottom row	Partition the 6.25U region	1U / 1.25U / 2.75U widths; stabilizers
Right Shift	Short Right Shift + additional 1U key	Split-Shift design practice
Backspace	Split if required	Multi-layout PCB practice
Enter	Use ISO-style physical region where appropriate	ISO keycap / plate practice
Control layer	Implement JIS keycodes and matrix	Existing MCU / firmware platforms

The PCB and plate remain JIS-specific where necessary. The engineering question is how much of the physical component and design system can be reused after compatibility is verified.

Programmable use beyond JIS

The physical value of these thumb keys is not necessarily limited to Japanese input. On QMK-based keyboards, for example, the same positions can support dual-role functions such as Mod-Tap. This suggests that the IC-JIS bottom row could be evaluated not only as a JIS-specific solution, but also as one variation of a broader programmable thumb-key architecture.

7 Engineering and Business Hypotheses to Validate



Reference only: ABNT2 illustrates that locale-specific keys can coexist with a broader physical-layout ecosystem. It is not a direct design basis for IC-JIS.

FUNCTION	HYPOTHESIS FOR PROTOTYPE / QUOTATION
Product planning	A JIS model can be evaluated as a localized variant on a common platform.
Mechanical / PCB	Split-space and split-Shift practices can reduce wholly unique design work.
Sourcing / manufacturing	Dependence on non-market dimensions, dedicated tooling, and dedicated inventory may be reduced.
Service	Replacement options may remain after original-part supply ends.
User experience	Japanese input functions can be retained while expanding repair and replacement options.

Quantification discipline

No cost reduction or development-time reduction is claimed at this stage. Compare tooling, unique part count, inventory units, assembly processes, and service-part supply against the current design, then decide from prototype results and quotations.

8 Validation Plan

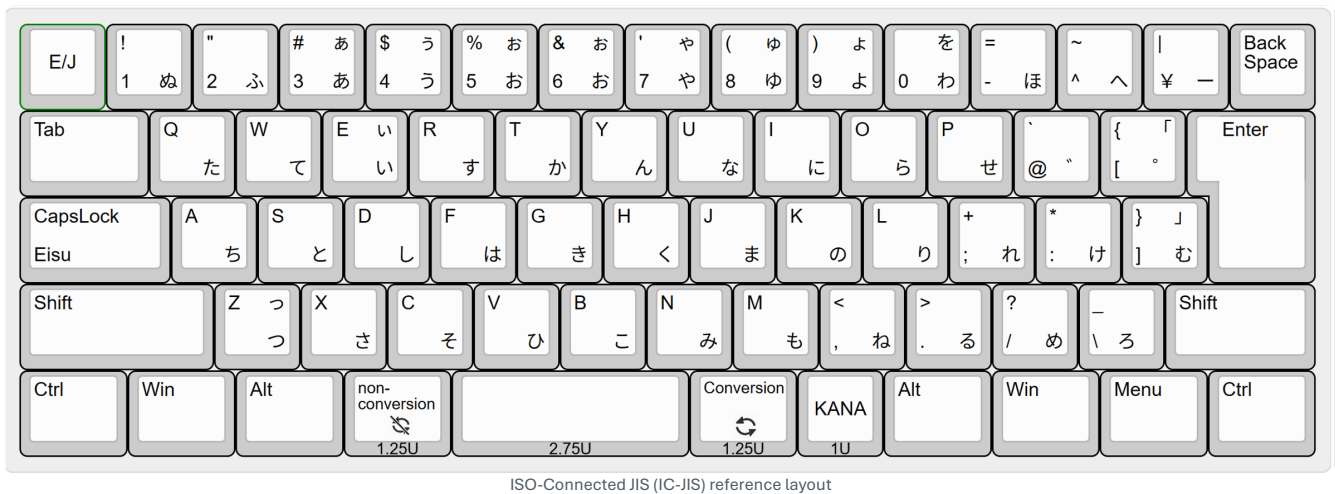
Adoption should not be decided from dimensions alone. Validation progresses from layout feasibility to prototype, usability, component comparison, and business evaluation.

STAGE	ACTIVITY	MEASURE / CHECK
Step 0	Desktop layout feasibility only	6.25U fit; switch/stabilizer positions; adjacent-key interference; scope of JIS-specific deltas
1. Engineering	Create PCB / plate / stabilizer layout	Mechanical fit; existing-part compatibility
2. Prototype	Build an IC-JIS sample	Rigidity; assembly effort; keycode operation
3. Usability	Run identical tasks on current JIS, long-Space and IC-JIS layouts	Input time; miskeys; adjacent-key confusion; thumb reachability
4. Components	Compare BOM and alternatives	Unique-part count; alternatives; stabilizer compatibility
5. Business	Estimate tooling, inventory, process and service supply	Total cost; commonization scope; supply continuity

Evaluation principles

- Evaluate romaji input, kana input, and direct IME control separately.
- Assess usability and long-term parts/service implications in the same decision framework.
- Define pass/fail criteria before prototyping; revise dimensions if the design does not satisfy them.

9 Recommended Next Action



FIRST DECISION

Do not decide whether to adopt IC-JIS. First decide whether to run Step 0: a desktop layout feasibility check with no commitment to new tooling, production PCB design, or product launch.

Step 0: desktop layout feasibility

- Fit 1.25U Non-Conversion + 2.75U Space + 1.25U Conversion + 1U Katakana/Hiragana inside the existing 6.25U region.
- Check 2.75U stem and stabilizer geometry against available parts or existing designs.
- Check interference and spacing with adjacent Alt / Win / Command positions.
- Check the “Ro” key using a short Right Shift + 1U split arrangement.
- Identify which PCB / plate elements can remain common and which genuinely require JIS-specific design.

Step 0 output

Deliverables: layout drawing, switch/stabilizer placement, list of common vs. JIS-specific deltas, and one decision: proceed, revise and re-check, or stop. If Step 0 shows physical feasibility, only then proceed to PCB/plate prototyping and comparative evaluation.

CONNECT, DON'T INVENT.

Keep JIS-specific functionality where it is needed. Reconnect everything else to reusable designs and components. IC-JIS is a starting point for testing that possibility.